

Linear Regression

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Math prerequisites for this lecture: You should know

- matrix notation, matrix-vector multiplication (Section II, Chapter 5 in Boyd and Vandenberghe)
- inner product/dot product (Section I, Chapter 1 in Boyd and Vandenberghe)
- derivatives and optimization (Appendix C in Boyd and Vandenberghe)
- norm of a vector (Section I, Chapter 3 in Boyd and Vandenberghe)
- matrix inverse (Section II, Chapter 11 in Boyd and Vandenberghe)

In this lecture

- Simple (univariate) linear regression
- Multiple linear regression
- Linear basis function regression
- OLS solution for simple regression
- OLS solution for multiple/LBF regression
- Interpretation

With linear regression, as with all of the supervised learning models in this course, we will consider:

- The parts of the basic “recipe” (loss function, training algorithm, etc.)

as well as:

- What type of relationships $f(x)$ can it represent?
- What insight can we get from the trained model?
- What is the cost of training/inference?
- How do we control the generalization error?

For linear regression, we will consider the first two questions in this lesson, and the second two questions in the next lesson.

Regression

Regression - quick review

The output variable y is continuously valued.

We need a function f to map each input vector $\mathbf{x}_i$ to a prediction,

$$\hat{y}_i = f(\mathbf{x}_i)$$

where (we hope!) $\hat{y}_i \approx y_i$.

Prediction by mean

We previously a simple model that predicts the mean of target variable in training data:

$$\hat{y}_i = w$$

$\forall i$, where $w = \frac{1}{N} \sum_{i=1}^N y_i = \bar{y}$.

We can show that the mean is the one-parameter model that optimizes the *mean squared error* (MSE) loss function:

$$L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Step 1: plug $\hat{y} = w$ into the loss function:

$$\text{minimize } \frac{1}{N} \sum (y - \hat{y})^2 \rightarrow \text{minimize } \frac{1}{N} \sum (y - w)^2$$

Step 2: take the derivative of the loss function with respect to the parameter w

$$\frac{\partial}{\partial w} \frac{1}{N} \sum (y - w)^2 = -\frac{2}{N} \sum (y - w)$$

Step 3: set the derivative equal to 0, and solve for the parameter w (L is convex, so this solution is a minimum):

$-\frac{2}{N} \sum (y - w) = 0$	Set the derivative equal to zero
$\sum (y) - \sum (w) = 0$	Multiply both sides by $-\frac{N}{2}$ and expand
$\sum (y) = \sum (w)$	Rearrange to get a sum on each side
$\sum (y) = Nw$	Sum the constant w over N samples
$\rightarrow w = \frac{1}{N} \sum (y)$	Solve for w

Prediction by mean, illustration

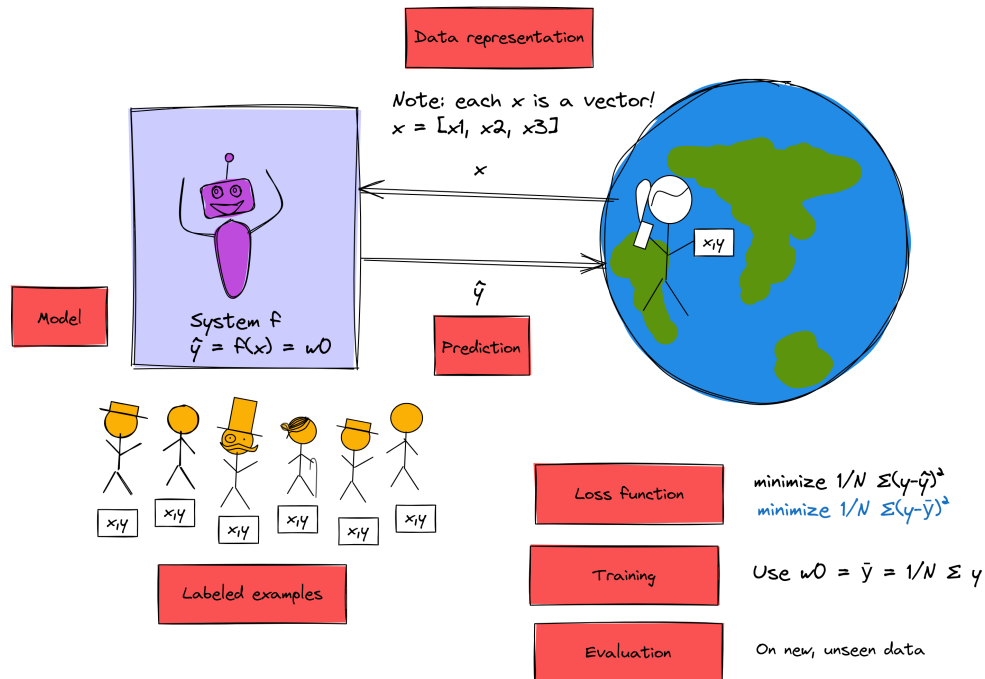


Figure 1: A “recipe” for our simple ML system.

Note that for “prediction by mean”, the loss function we defined for this problem - sum of squared differences between the true value and predicted value:

$$\frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^2$$

is the variance of y .

Under what conditions will that loss function be very small (or even zero)?

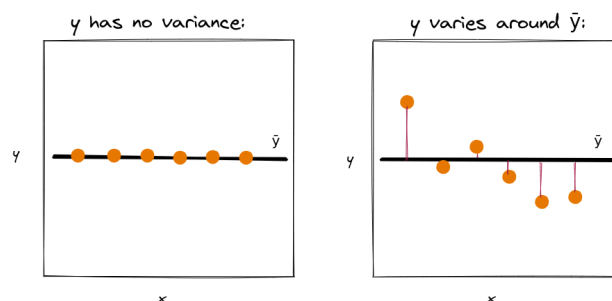


Figure 2: When is prediction by mean good enough?

Prediction by mean is a good model if there is no *variance* in y . But, if there is variance in y , a good model should *explain* some/all of that variance.

Mean, variance, covariance - definitions

Mean, variance, covariance:

$$\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i, \quad \sigma_y^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^2$$
$$\sigma_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$

(We are using the “biased” estimates, without Bessel’s correction.)

Simple linear regression

A “simple” linear regression is a linear regression with only one feature.

Regression with one feature

For simple linear regression, we have feature-label pairs:

$$(x_i, y_i), i = 1, 2, \dots, N$$

(we’ll often drop the index i when it’s convenient.)

Simple linear regression model

Assume a linear relationship:

$$\hat{y}_i = w_0 + w_1 x_i$$

where $\mathbf{w} = [w_0, w_1]$, the intercept and slope, are model **parameters** that we *fit* in training.

Residual term (1)

There is variance in y among the data:

- some of it is “explained” by $f(x) = w_0 + w_1 x$
- some of the variance in y is *not* explained by $f(x)$

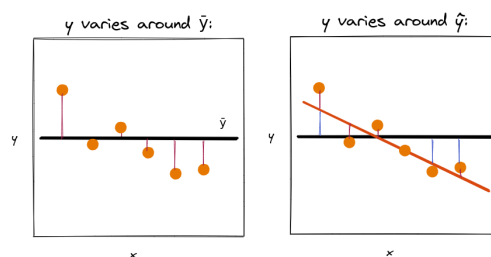


Figure 3: Some (but) not necessarily all of variance of y is explained by the linear model.

Maybe y varies with some other function of x , maybe part of the variance in y is explained by other features not in x , maybe it is truly “random”...

Residual term (2)

The *residual* term captures everything that isn't in the model:

$$y_i = w_0 + w_1 x_i + e_i$$

where $e_i = y_i - \hat{y}_i$.

Example: Intro ML grades

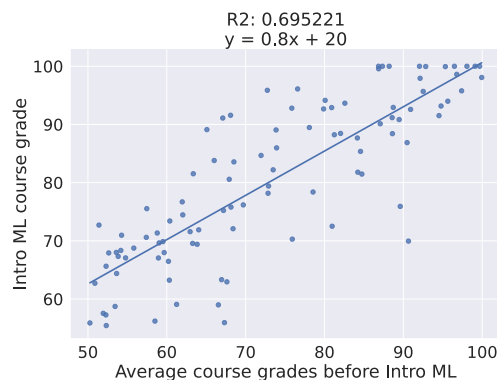


Figure 4: Predicting students' grades in Intro ML using regression on previous coursework.

Suppose students we want to develop a model that can predict a student's course grade.

To some extent, a student's average grades on previous coursework "explains" their grade in Intro ML.

- The predicted value for each student, $\hat{y}$, is along the diagonal line. Draw a vertical line from each student's point (y) to the corresponding point on the line ($\hat{y}$). This is the residual $e = y - \hat{y}$.
- Some students fall right on the line - these are examples that are explained "well" by the model.
- Some students are far from the line. The magnitude of the *residual* is greater for these examples.
- The difference between the "true" value y and the predicted value $\hat{y}$ may be due to all kinds of differences between the "well-explained example" and the "not-well-explained-example" - not everything about Intro ML course grade can be explained by performance in previous coursework! This is what the residual captures.

Interpreting the linear regression: If slope w_1 is 0.8 points in Intro ML per point average in previous coursework, we can say that

- a 1-point increase in score on previous coursework is, on average, associated with a 0.8 point increase in Intro ML course grade.

What can we say about possible explanations? We can't say much using this method - anything is possible:

- statistical fluke (we haven't done any test for significance)
- causal - students who did well in previous coursework are better prepared
- confounding variable - students who did well in previous coursework might have more time to study because they don't have any other jobs or obligations, and they are likely to do well in Intro ML for the same reason.

This method doesn't tell us *why* this association is observed, only that it is. (There are other methods in statistics for determining whether it is a statistical fluke, or for determining whether it is a causal relationship.)

(Also note that the 0.8 point increase according to the regression model is only an *estimate* of the "true" relationship.)

Ordinary least squares solution for simple linear regression

Solving simple linear regression (1)

Step 1: plug the model into the MSE loss function:

$$L(w_0, w_1) = \frac{1}{N} \sum_{i=1}^N [y_i - (w_0 + w_1 x_i)]^2$$

Solving simple linear regression (2)

Step 2: take the partial derivatives with respect to the parameters:

$$\begin{aligned}\frac{\partial L}{\partial w_0} &= -\frac{2}{N} \sum_{i=1}^N [y_i - (w_0 + w_1 x_i)] \\ \frac{\partial L}{\partial w_1} &= -\frac{2}{N} \sum_{i=1}^N x_i (y_i - w_0 - w_1 x_i).\end{aligned}$$

Solving simple linear regression (3)

Step 3: set the derivatives equal to zero and solve (L is convex, so this is a minimum):

$$\begin{aligned}\frac{\partial L}{\partial w_0} = 0 &\implies w_0^* = \bar{y} - w_1^* \bar{x} \\ \frac{\partial L}{\partial w_1} = 0 &\implies w_1^* = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}.\end{aligned}$$

Assume the x_i values are not all equal, so $\sum_{i=1}^N (x_i - \bar{x})^2 > 0$.

First, use the derivative with respect to w_0 to solve for the intercept:

$$-\frac{2}{N} \sum_{i=1}^N [y_i - (w_0 + w_1 x_i)] = 0 \quad \text{Set the derivative equal to zero}$$

$$\sum_{i=1}^N [y_i - (w_0 + w_1 x_i)] = 0 \quad \text{Multiply both sides by } -\frac{N}{2}$$

$$\sum_{i=1}^N y_i - Nw_0 - w_1 \sum_{i=1}^N x_i = 0 \quad \text{Distribute the sum}$$

$$N\bar{y} - Nw_0 - Nw_1 \bar{x} = 0 \quad \text{Use } \sum y_i = N\bar{y} \text{ and } \sum x_i = N\bar{x}$$

$$w_0 = \bar{y} - w_1 \bar{x} \quad \text{Divide by } N \text{ and solve for } w_0$$

$$\therefore w_0^* = \bar{y} - w_1^* \bar{x} \quad \text{Identify the minimizing coefficients}$$

Next, use the derivative with respect to w_1 to solve for the slope:

$$-\frac{2}{N} \sum_{i=1}^N x_i (y_i - w_0 - w_1 x_i) = 0$$

Set the derivative equal to zero

$$\sum_{i=1}^N x_i (y_i - w_0 - w_1 x_i) = 0$$

Multiply both sides by $-\frac{N}{2}$

$$\sum_{i=1}^N x_i y_i - w_0 \sum_{i=1}^N x_i - w_1 \sum_{i=1}^N x_i^2 = 0$$

Distribute x_i and the sum

$$\sum_{i=1}^N x_i y_i - (\bar{y} - w_1 \bar{x}) N \bar{x} - w_1 \sum_{i=1}^N x_i^2 = 0$$

Substitute $w_0 = \bar{y} - w_1 \bar{x}$

$$w_1 \left(\sum_{i=1}^N x_i^2 - N \bar{x}^2 \right) = \sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}$$

Collect the terms containing w_1

$$w_1 = \frac{\sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}}{\sum_{i=1}^N x_i^2 - N \bar{x}^2}$$

Divide by the coefficient of w_1

The numerator and denominator can be written as:

$$\begin{aligned} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) &= \sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}, \\ \sum_{i=1}^N (x_i - \bar{x})^2 &= \sum_{i=1}^N x_i^2 - N \bar{x}^2. \end{aligned}$$

Therefore,

$$w_1^* = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}.$$

and note that this is a ratio of covariance of x and y , to variance of x :

$$w_1^* = \frac{\sigma_{xy}}{\sigma_x^2}$$

We can also express it in terms of correlation coefficient $r_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$:

$$w_1^* = \frac{r_{xy} \sigma_y}{\sigma_x}$$

(Note: from Cauchy-Schwartz law, $|\sigma_{xy}| < \sigma_x \sigma_y$, we know $r_{xy} \in [-1, 1]$)

Understanding correlation coefficient

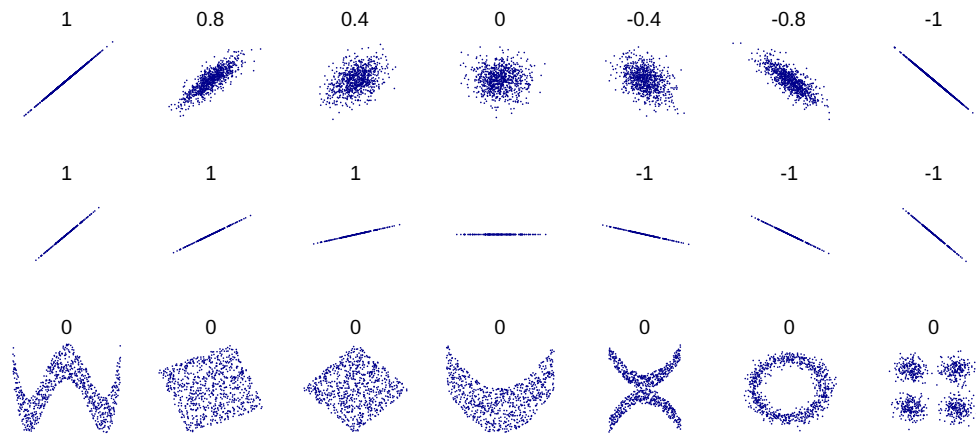


Figure 5: Several sets of (x, y) points, with r_{xy} for each. Image via Wikipedia.

The correlation coefficient $\frac{\sigma_{xy}}{\sigma_x \sigma_y}$ is fundamental to the data - it is not about a fitted model. When we say

$$w_1^* = \frac{r_{xy} \sigma_y}{\sigma_x}$$

we mean, the *optimal* parameter has this relationship to the fundamental structure in the data.

Understanding the coefficient

- What does w_0 do to the line?
- What does w_1 do to the line?

“an increase of one unit in this feature is associated with an increase of the target variable by w_1 ”

Doesn't tell us about causality, significance, etc.!

Interpreting regression metrics

Interpreting MSE, MAE

Mean Squared Error (MSE):

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Mean Absolute Error (MAE):

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

MSE is useful for training because:

- The squared loss function is differentiable everywhere, making it easy to optimize
- It has a closed-form solution (the normal equations)
- It penalizes large errors more heavily

However, MSE is less intuitive for *interpreting* model error:

- MSE is in squared units of the target variable, hard to understand in context
- MAE is in the same units as the target variable
- MAE is more robust to outliers

If y is in dollars, you can directly say “on average, my predictions are off by \$X” using MAE. (We could also use RMSE - take the square root of the MSE, but RMSE is also sensitive to outliers.)

In some cases, we may prefer Mean Absolute Percent Error:

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

For example, suppose you are predicting house price. A \$50k prediction error is much more significant for the 200k home (25% off) than for a 5 million dollar one (1% off). MAPE calculates error as a percentage of each actual value, so you get a meaningful comparison across different price ranges.

Interpreting R2 as explained variance

$$R^2 = 1 - \frac{MSE}{\sigma_y^2} = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y}_i)^2}$$

For linear regression: What proportion of the variance in y is “explained” by our model?

- $R^2 \approx 1$ - model “explains” all the variance in y
- $R^2 \approx 0$ - model doesn’t “explain” any of the variance in y

Interpreting R2 as error relative to “mean model”

Alternatively: what is the ratio of error of our model, to error of prediction by mean?

$$R^2 = 1 - \frac{MSE}{\sigma_y^2} = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y}_i)^2}$$

What would be R^2 of a model that is *worse* than prediction by mean?

Example: Intro ML grades (2)

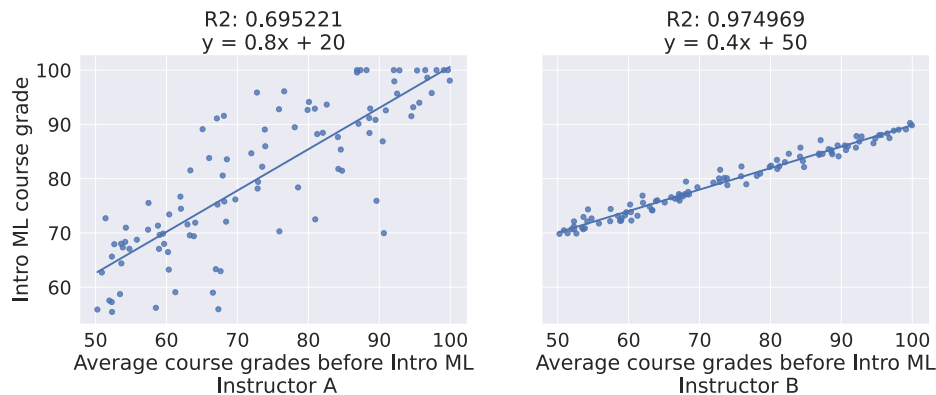


Figure 6: Predicting students’ grades in Intro ML, for two different sections.

In Instructor A’s section, a change in average overall course grades is associated with a bigger change in Intro ML course grade than in Instructor B’s section; but in Instructor B’s section, more of the variance among students is explained by the linear regression on previous overall grades.

Multiple linear regression

Matrix representation of data

Represent data as a **matrix**, with N samples and d features; one sample per row and one feature per column:

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,d} \\ \vdots & \ddots & \vdots \\ x_{N,1} & \cdots & x_{N,d} \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

Thus, $\mathbf{X} \in \mathbb{R}^{N \times d}$ and $\mathbf{y} \in \mathbb{R}^N$.

$x_{i,j}$ is j th feature of i th sample.

Note: by convention, we use capital letter for matrix, bold lowercase letter for vector.

Linear model

For a given sample (row), assume a linear relationship between feature vector $\mathbf{x}_i = [x_{i,1}, \dots, x_{i,d}]$ and scalar target variable y_i :

$$\hat{y}_i = w_0 + w_1 x_{i,1} + \cdots + w_d x_{i,d}$$

Model has $d + 1$ **parameters**.

- Samples are vector-label pairs: $(\mathbf{x}_i, y_i), i = 1, 2, \dots, N$
- Each sample has a feature vector $\mathbf{x}_i = [x_{i,1}, \dots, x_{i,d}]$ and scalar target y_i
- Predicted value for i th sample will be $\hat{y}_i = w_0 + w_1 x_{i,1} + \cdots + w_d x_{i,d}$

It's a little awkward to carry around that w_0 separately, if we roll it in to the rest of the weights we can use a matrix representation...

Matrix representation of linear regression (1)

Define a new **feature matrix** and **weight vector**:

$$\mathbf{A} = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & \cdots & x_{N,d} \end{bmatrix}, \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}$$

Thus, $\mathbf{A} \in \mathbb{R}^{N \times (d+1)}$ and $\mathbf{w} \in \mathbb{R}^{d+1}$.

Matrix representation of linear regression (2)

Then, $\hat{\mathbf{y}} = \mathbf{A}\mathbf{w}$.

For a new sample, define $\tilde{\mathbf{x}}_i = [1, \mathbf{x}_i^T]^T \in \mathbb{R}^{d+1}$. Its predicted value is $\hat{y}_i = \tilde{\mathbf{x}}_i^T \mathbf{w}$.

(The angle brackets denote a dot product.)

Here is an example showing the computation:

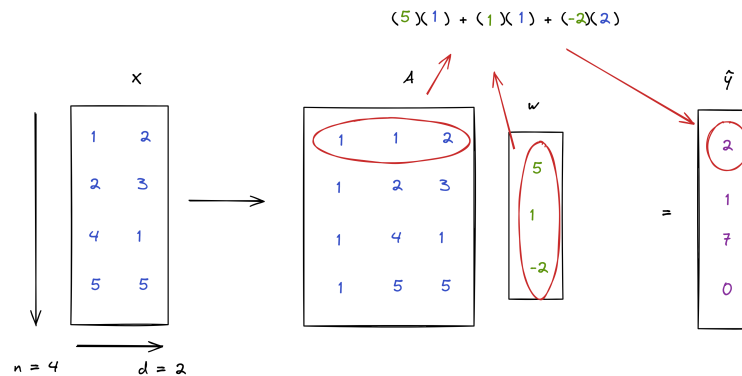


Figure 7: Example of a multiple linear regression.

What does the residual look like in the multivariate case?

Illustration - multiple linear regression

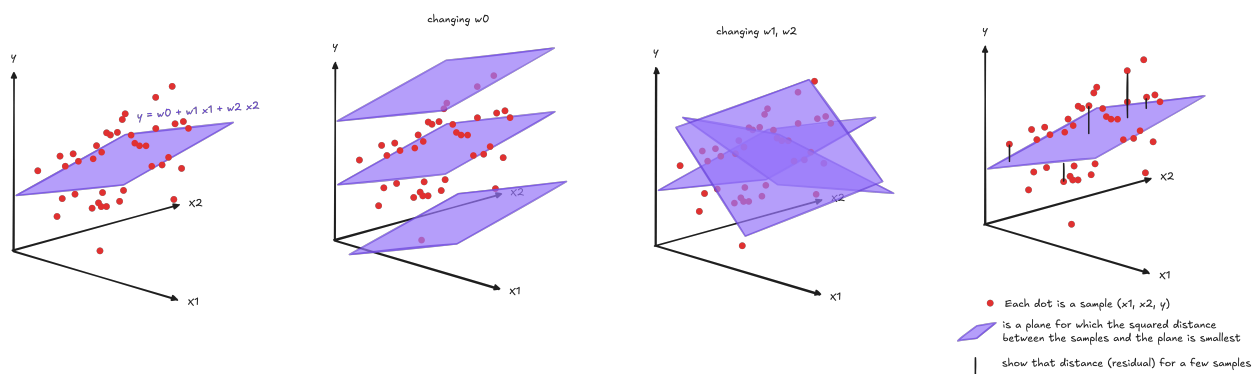


Figure 8: In 2D, the least squares regression is now a plane. In higher d , it's a hyperplane.

Understanding the coefficients - multiple regression

The coefficient w_j for feature j says:

- simple regression: "an increase of one unit in this feature is associated with an increase of the target variable by w_j "
- multiple regression: "an increase of one unit in this feature, **while holding the other features that are in the model constant**, is associated with an increase of the target variable by w_j "

Note: doesn't say whether the effect is *causal* or whether it is *significant* (out of scope of this course).

Be aware of units - we cannot directly compare the magnitude of coefficients of features measured in different units.

Also, be aware of how coefficient of one feature can relate to coefficient of another:

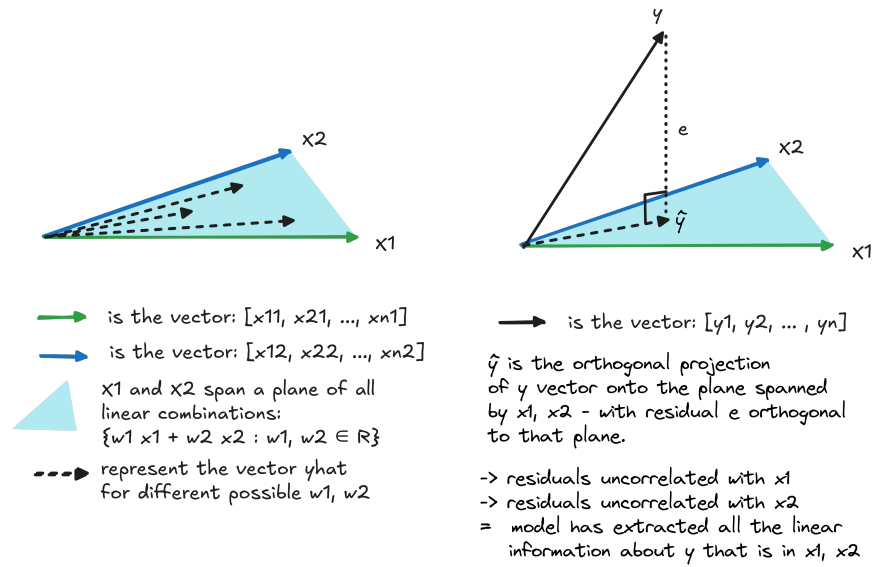


Figure 9: Representation of features and predictions as vectors.

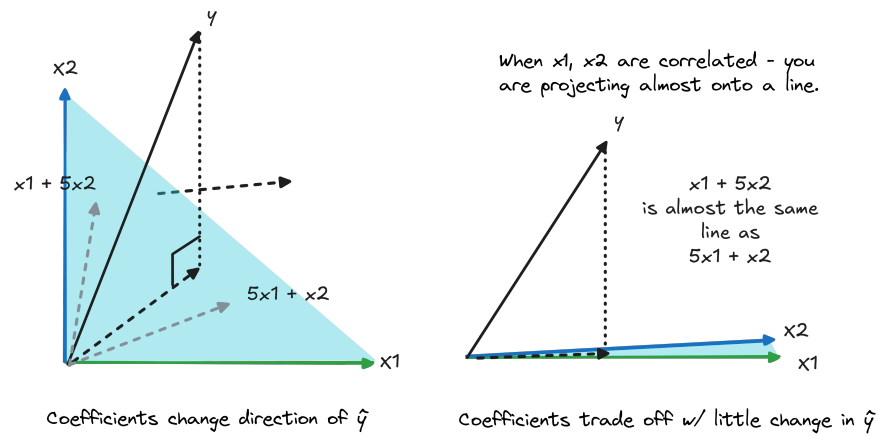
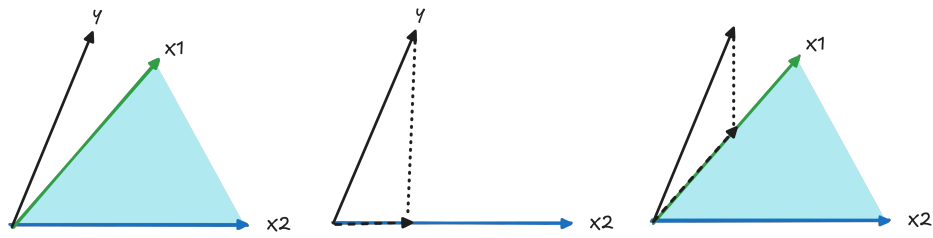


Figure 10: Meaning of coefficients when features are collinear.



Imagine there is a hidden confounding variable x_h that affects both x_1 and x_2 in similar ways, making them correlated. but only x_1 is related to y directly.

e.g. x_h = conscientiousness
 x_1 = time spent studying ML
 x_2 = time spent studying for other courses
 y = ML course grade

Figure 11: Meaning of coefficients when there is a hidden confounding variable.

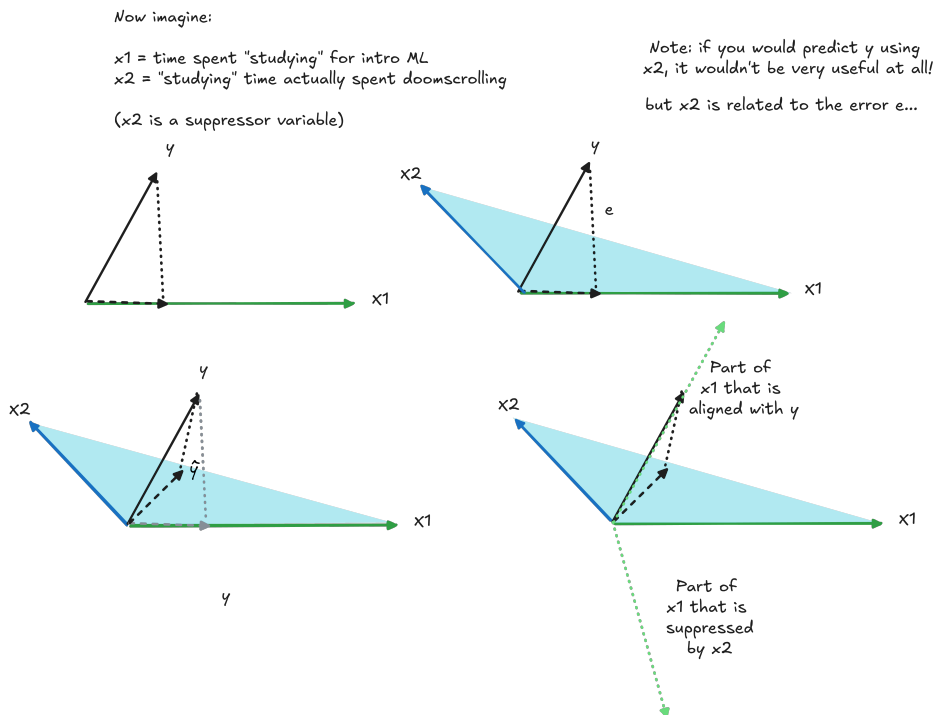


Figure 12: Meaning of coefficients with suppressor variable.

Linear basis function regression

The assumption that the output is a linear function of the input features is very restrictive. Instead, what if we consider *linear combinations of fixed non-linear functions*?

Basis functions

Define a **basis function**:

$$\phi_j(\mathbf{x}) = \phi_j(x_1, \dots, x_d)$$

that “transforms” the original features.

Linear basis function model for regression

Standard linear model:

$$\hat{y}_i = w_0 + w_1 x_{i,1} + \dots + w_d x_{i,d}$$

Linear basis function model:

$$\hat{y}_i = w_0 \phi_0(\mathbf{x}_i) + \dots + w_p \phi_p(\mathbf{x}_i)$$

Some notes:

- The 1s column we added to the design matrix is easily represented as a basis function ($\phi_0(\mathbf{x}) = 1$).
- There is not necessarily a one-to-one correspondence between the columns of X and the basis functions ($p \neq d$ is OK!). You can have more/fewer basis functions than columns of X .
- Each basis function can accept as input the entire vector $\mathbf{x}_i$.
- The model has $p + 1$ parameters.

Vector form of linear basis function model

The prediction of this model for one sample i , expressed in vector form, is:

$$\hat{y}_i = \langle \phi(\mathbf{x}_i), \mathbf{w} \rangle = \mathbf{w}^T \phi(\mathbf{x}_i)$$

where

$$\phi(\mathbf{x}_i) = [\phi_0(\mathbf{x}_i), \dots, \phi_p(\mathbf{x}_i)], \mathbf{w} = [w_0, \dots, w_p]$$

(The angle brackets denote a dot product.)

Important note: although the model can be non-linear in $\mathbf{x}$, it is still linear in the parameters $\mathbf{w}$ (note that $\mathbf{w}$ appears *outside* $\phi(\cdot)$!) That’s what makes it a *linear model*.

Some basis functions have their own parameters that appear inside the basis function, i.e. we might have a model

$$\hat{y}_i = \mathbf{w}^T \phi(\mathbf{x}_i, \boldsymbol{\theta})$$

where $\boldsymbol{\theta}$ are the parameters of the basis function. The model is *non-linear* in those parameters, and they need to be fixed before training.

Matrix form of linear basis function model

Given data $(\mathbf{x}_i, y_i), i = 1, \dots, N$:

$$\Phi = \begin{bmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_p(\mathbf{x}_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_p(\mathbf{x}_N) \end{bmatrix}$$

and $\hat{\mathbf{y}} = \Phi \mathbf{w}$.

Here, $\Phi \in \mathbb{R}^{N \times (p+1)}$ and $\mathbf{w} \in \mathbb{R}^{p+1}$ because the columns run from ϕ_0 through ϕ_p .

Ordinary least squares solution for multiple/basis function regression

Least squares loss for multiple/LBF regression

Given $\mathbf{y} \in \mathbb{R}^N$ and $\Phi \in \mathbb{R}^{N \times (p+1)}$, we'll use loss function

$$L(\mathbf{w}) = \frac{1}{2} \|\mathbf{y} - \hat{\mathbf{y}}\|^2$$

where the norm above is the L2 norm.

(we defined it with a $\frac{1}{2}$ constant factor for convenience.)

Setup: L2 norm

Definition: L2 norm of a vector $\mathbf{x} = (x_1, \dots, x_n)$:

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2}$$

We will want to minimize the L2 norm of the residual. (Equivalent to minimizing squared error!)

Setup: Gradient vector

To minimize a multivariate function $f(\mathbf{x}) = f(x_1, \dots, x_n)$, we find places where the **gradient** is zero, i.e. each entry must be zero:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix}$$

The gradient is the vector of partial derivatives.

Solving multiple/LBF regression (1)

Step 1: plug $\hat{\mathbf{y}} = \Phi \mathbf{w}$ into the loss function:

$$\text{minimize } \frac{1}{2} \|\mathbf{y} - \hat{\mathbf{y}}\|^2 \rightarrow \text{minimize } \frac{1}{2} \|\mathbf{y} - \Phi \mathbf{w}\|^2$$

Solving multiple/LBF regression (2)

Step 2: take the gradient of the loss function with respect to the parameter vector $\mathbf{w}$:

$$\nabla L(\mathbf{w}) = -\Phi^T(\mathbf{y} - \Phi \mathbf{w})$$

For those interested in the detailed derivation:

First, expand the squared norm. Transposing a product reverses the order of its factors, so $(\Phi \mathbf{w})^T = \mathbf{w}^T \Phi^T$.

$$\begin{aligned}
L(\mathbf{w}) &= \frac{1}{2}(\mathbf{y} - \Phi \mathbf{w})^T(\mathbf{y} - \Phi \mathbf{w}) && \text{Use } \|\mathbf{a}\|^2 = \mathbf{a}^T \mathbf{a} \\
&= \frac{1}{2}(\mathbf{y}^T - \mathbf{w}^T \Phi^T)(\mathbf{y} - \Phi \mathbf{w}) && \text{Transpose the first factor} \\
&= \frac{1}{2}(\mathbf{y}^T \mathbf{y} - \mathbf{y}^T \Phi \mathbf{w} - \mathbf{w}^T \Phi^T \mathbf{y} + \mathbf{w}^T \Phi^T \Phi \mathbf{w}) && \text{Distribute without changing factor order} \\
&= \frac{1}{2}\mathbf{y}^T \mathbf{y} - \mathbf{w}^T \Phi^T \mathbf{y} + \frac{1}{2}\mathbf{w}^T \Phi^T \Phi \mathbf{w} && \text{Combine the equal scalar cross terms}
\end{aligned}$$

The cross terms are equal because a scalar equals its transpose:

$$\mathbf{y}^T \Phi \mathbf{w} = (\mathbf{y}^T \Phi \mathbf{w})^T = \mathbf{w}^T \Phi^T \mathbf{y}.$$

Now take the gradient of each term. We use $\nabla_{\mathbf{w}} c = 0$, $\nabla_{\mathbf{w}}(\mathbf{w}^T \mathbf{a}) = \mathbf{a}$, and $\nabla_{\mathbf{w}}(\frac{1}{2}\mathbf{w}^T A \mathbf{w}) = \frac{1}{2}(A + A^T)\mathbf{w}$.

$$\begin{aligned}
\nabla_{\mathbf{w}} \left(\frac{1}{2} \mathbf{y}^T \mathbf{y} \right) &= 0 && \mathbf{y} \text{ does not depend on } \mathbf{w} \\
\nabla_{\mathbf{w}} (-\mathbf{w}^T \Phi^T \mathbf{y}) &= -\Phi^T \mathbf{y} && \text{Use the linear-term rule} \\
\nabla_{\mathbf{w}} \left(\frac{1}{2} \mathbf{w}^T \Phi^T \Phi \mathbf{w} \right) &= \Phi^T \Phi \mathbf{w} && \Phi^T \Phi \text{ is symmetric} \\
\nabla L(\mathbf{w}) &= -\Phi^T \mathbf{y} + \Phi^T \Phi \mathbf{w} && \text{Add the three gradients} \\
&= -\Phi^T(\mathbf{y} - \Phi \mathbf{w}) && \text{Factor out } -\Phi^T
\end{aligned}$$

Solving multiple/LBF regression (3)

Step 3: set the gradient equal to 0, and solve for the parameter vector $\mathbf{w}$. We find:

$$\mathbf{w}^* = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

Assume Φ has full column rank, so $\Phi^T \Phi$ is invertible. Because matrix multiplication is not commutative, we must multiply both sides from the left.

$$-\Phi^T(\mathbf{y} - \Phi \mathbf{w}) = 0$$

Set the gradient equal to zero

$$\Phi^T(\mathbf{y} - \Phi \mathbf{w}) = 0$$

Multiply both sides by -1

$$\Phi^T \mathbf{y} - \Phi^T \Phi \mathbf{w} = 0$$

Distribute Φ^T over the subtraction

$$\Phi^T \Phi \mathbf{w} = \Phi^T \mathbf{y}$$

Add $\Phi^T \Phi \mathbf{w}$ to both sides

$$(\Phi^T \Phi)^{-1} \Phi^T \Phi \mathbf{w} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

Left-multiply by $(\Phi^T \Phi)^{-1}$

$$I \mathbf{w} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

Use $(\Phi^T \Phi)^{-1}(\Phi^T \Phi) = I$

$$\mathbf{w} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

Use $I \mathbf{w} = \mathbf{w}$

$$\therefore \mathbf{w}^* = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

Identify the minimizing parameter vector

Solving a set of linear equations

If Φ has full column rank $p + 1$ (which requires $N \geq p + 1$), then the solution is unique:

$$\mathbf{w}^* = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

This expression:

$$\Phi^T \Phi \mathbf{w} = \Phi^T \mathbf{y}$$

represents a set of $p + 1$ equations in $p + 1$ unknowns, called the *normal equations*.

We can solve this as we would any set of linear equations (see supplementary notebook on computing regression coefficients by hand.)

Recap

Completed “recipe”

1. Get **data**: $(\mathbf{x}_i, y_i), i = 1, 2, \dots, N$
2. Choose a **model**: $\hat{y}_i = \langle \phi(\mathbf{x}_i), \mathbf{w} \rangle$
3. Choose a **loss function**: $L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$
4. Find model **parameters** that minimize loss: OLS solution for $\mathbf{w}^*$
5. Use model to **predict** $\hat{y}$ for new, unlabeled samples
6. Evaluate model performance on new, unseen data

Key questions

- What type of relationships $f(x)$ can it represent?
- What insight can we get from the trained model?
- (What is the cost of training/inference?)
- (How do we control the generalization error?)

We will address the last two questions next week.